~ Internship Technical Report ~

DAY 01

Advanced TCP Port Scanner

Internship Record | VIEH GROUP

Name	Mubariz Mohd Ahmed Shaikh
Internship Program	VIEH – 30-Day Cybersecurity Automation Series
Task Title	Advanced TCP Port Scanner
Tools & Technologies	Python, Socket Programming, Threading
Operating Platform	Kali Linux

✿ Project Goal

The aim of this project is to develop an Advanced TCP Port Scanner using Python. This utility examines a designated host to discover open TCP ports and recognize active services. Additional capabilities — including concurrent threading, banner extraction, service identification, and exporting scan output to a file — were incorporated to boost both performance and overall utility.

✿ Overview

The Advanced TCP Port Scanner is a Python-driven cybersecurity utility designed to examine a target machine for open TCP ports. The primary goal is to gain insight into how TCP communication operates and how network services can be mapped through socket programming. This tool supports fundamental network reconnaissance and security auditing by detecting accessible ports and pinpointing the services running on them.

Throughout this project, Python socket programming was leveraged to initiate TCP connections with target ports. Concurrent threading was employed to enhance scanning efficiency by evaluating multiple ports simultaneously. Features such as service identification, banner extraction, and storing scan output to a text file were added to further enrich the tool's capabilities.

The scanner effectively locates open ports, reveals the active service, attempts to capture service banners, and stores all findings in a dedicated report file. This project offered hands-on exposure to networking concepts, Python automation, multi-threading techniques, and cybersecurity reconnaissance methods.

✿ Core Capabilities

- ❀ Rapid TCP Port Scanning
- ❀ Concurrent Multi-threaded Scanning
- ❀ Automatic Service Identification
- ❀ Banner Extraction from Active Services
- ❀ Export Findings to a Text File
- ❀ Flexible Target Configuration (IP or Domain)
- ❀ Beginner-Accessible Design
- ❀ Minimal Footprint & Easy to Deploy

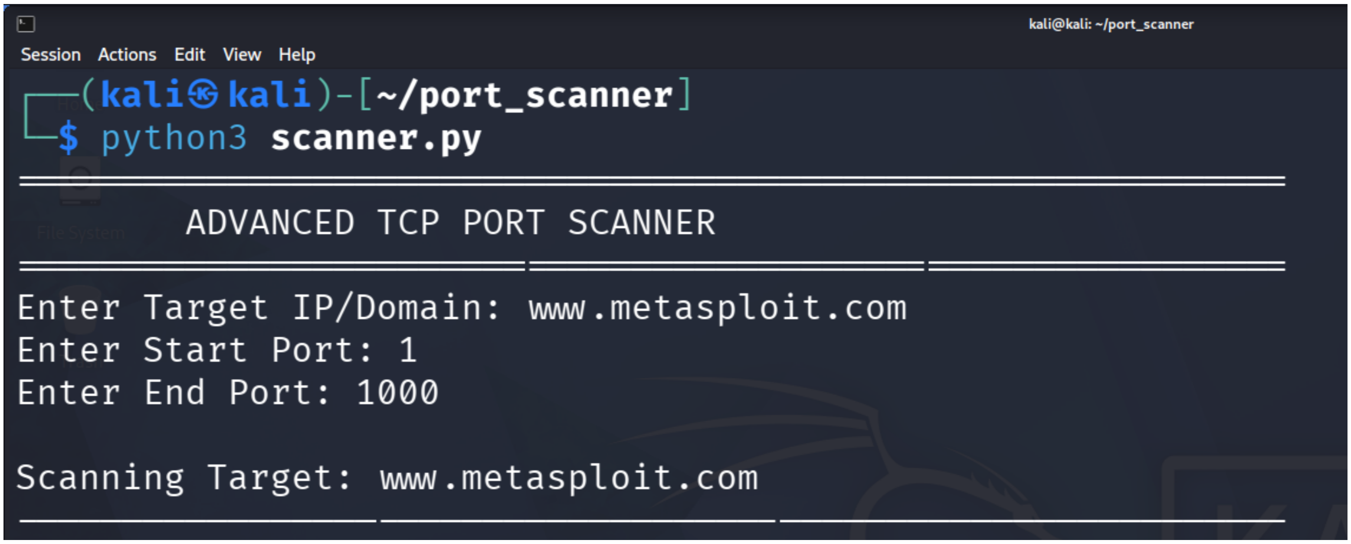
✿ Tech Stack

- ❀ Python 3
- ❀ Socket Programming
- ❀ Threading Module

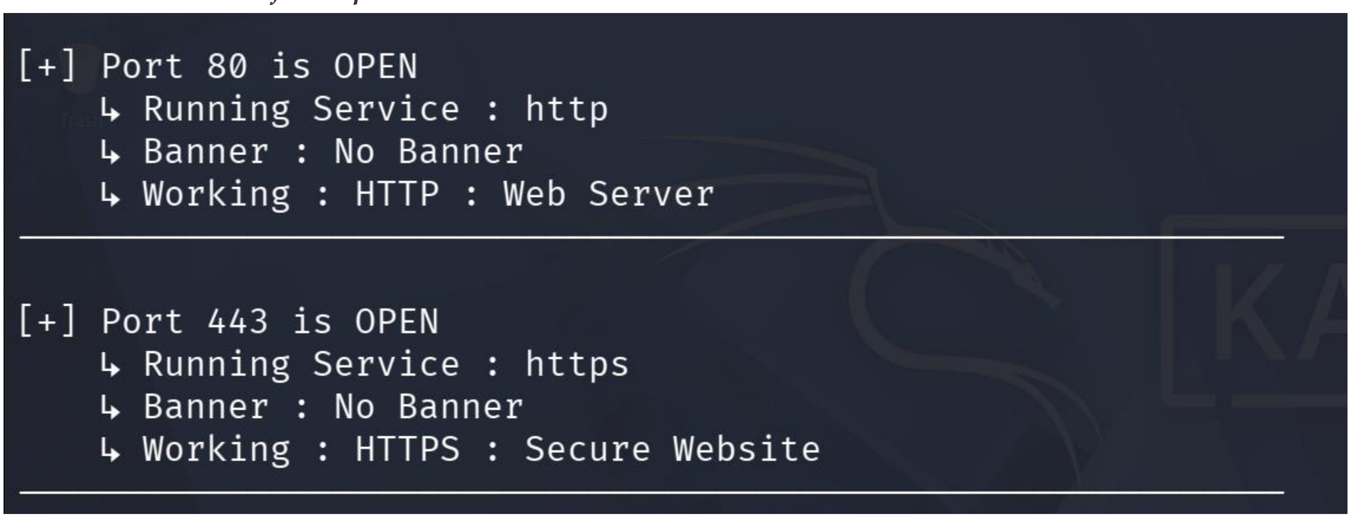
- Queue Module
- Kali Linux

Visual Demonstrations

Screenshot 1 – Launching the Scanner



Screenshot 2 – Identified Open Ports



Screenshot 3 – scan_results.txt File Output

```
────────────────────────────────────────────────────────────

[√] Scan Completed.
[√] Results exported to scan_results.txt
```

✿ How It Operates

- The user supplies the target IP/domain, start port, and end port.
- The program resolves the provided domain to its corresponding IP address.
- Multiple threads are spawned to accelerate the scanning process.
- Each port is evaluated via individual TCP socket connections.
- For any open port discovered, the service name is identified, banner data is retrieved, and a brief port description is displayed.
- All scan findings are consolidated and saved toscan_results.txt.

✿ Sample Output

Port	Details
22 (SSH)	[+] Status: OPEN | Service: ssh | Banner: OpenSSH | Description: Secure Remote Login
80 (HTTP)	[+] Status: OPEN | Service: http | Banner: Apache | Description: HTTP Web Server

✿ Key Takeaways

- Fundamentals of Socket Programming
- TCP Communication Concepts
- Multi-threading Techniques in Python
- Network Reconnaissance & Enumeration
- Service Identification Methods
- Banner Extraction Procedures
- File Handling & I/O in Python
- Cybersecurity Automation Essentials

✿ Final Thoughts

This project deepened understanding of how port scanners function and how network services can be discovered through Python programming. The integration of concurrent threading greatly improved scanning speed, while banner extraction and service identification provided richer insight into the target environment. Overall, this project reinforced practical expertise in networking fundamentals, Python automation, and foundational cybersecurity practices.

❀ ✿ ❀ ✿ ❀

⚠ Disclaimer

This tool is developed exclusively for academic study and authorized security evaluations. Performing unauthorized scans on any system without explicit permission may constitute a legal offence.

❀ ❀ ✿ ❁ ✿ ❀ ❀